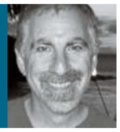
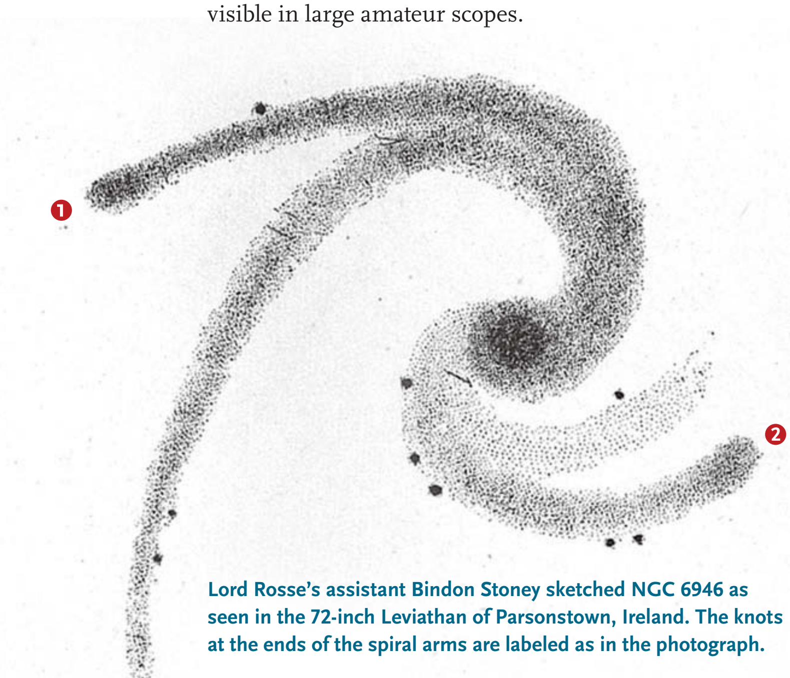


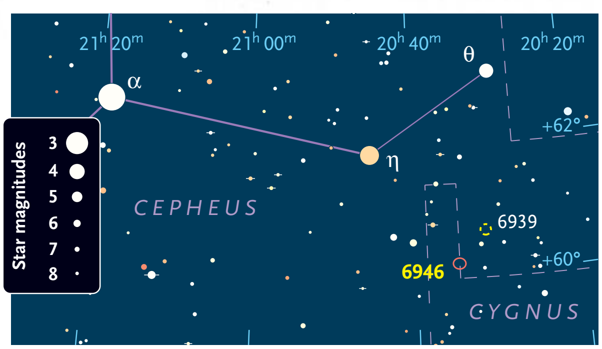
Unraveling NGC 6946

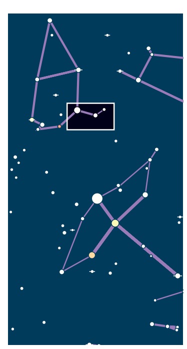
This far-northern galaxy's spiral arms are unusually easy to resolve.

THE SPIRAL GALAXY NGC 6946 is the most productive supernova factory of the past century. Since the last Milky Way supernova explosion 140 years ago, NGC 6946 has churned out nine supernovae — in 1917, 1939, 1948, 1968, 1969, 1980, 2002, 2004, and 2008 — earning it the nickname "Firecracker Galaxy." Hyperactive star formation runs rampant throughout the galaxy, from an inner nuclear starburst region to the outer disk, even producing an extraordinary super-star cluster (SSC) that's visible in large amateur scopes.



Lord Rosse's assistant Bindon Stoney sketched NGC 6946 as seen in the 72-inch Leviathan of Parsonstown, Ireland. The knots at the ends of the spiral arms are labeled as in the photograph.





At declination +60°, NGC 6946 lies far enough north to be circumpolar for most of the United States. It's especially well placed during the summer and autumn for northern observers to unravel its spiral arms.

NGC 6946 shines at magnitude 8.8 in a glittering star field at the border of Cygnus and Cepheus, about 2° southwest of 3.4-magnitude Eta (η) Cephei. It shares a low-power field with the 8th-magnitude open cluster NGC 6939. But at a distance of 15 to 20 million light years, the galaxy is roughly 4,000 times farther than its eyepiece neighbor. This odd couple is fairly easy to spot in my 15×50 binoculars. Both objects appear similar in size, though NGC 6939 displays a higher surface brightness.

Due to its location just 11° from the galactic plane, we view NGC 6946 through a veil of Milky Way dust. But because this late Hubble-type Scd spiral is relatively nearby and oriented nearly face-on to our line of sight, its spiral arms are unusually easy to resolve. In the fall of 1850, Irish astronomer William Parsons (Third Earl of Rosse) turned his 72-inch speculum-metal reflector toward NGC 6946 and found a "new spiral, very fine but faint; 3 branches, of which two terminate in knots, a fourth branch north preceding very doubtful." Later observations confirmed the fourth arm, and a sketch by observing assistant Bindon Stoney (shown at left) captures the asymmetric arm structure as well as the two knots.

Under dark skies, an 8-inch telescope should reveal the brightest arm, and a 12- to 14-inch should show the overall spiral structure. In the early 1980s, I was mesmerized when my 13-inch displayed the two prominent arms curling around the east side as well as a diffuse western arm.

Through my 18-inch Dobsonian, the irregular halo of NGC 6946 spans 8′ and grows brighter to a 1.5′ central region. The nucleus, however, is just a very small, weak enhancement, pierced by a faint starlike point. The dominant arm is attached at the north side of the core and unfurls counterclockwise to the east, passing just south of a 13.5-magnitude star before terminating at a 10″ H II knot (labeled **1** on the photograph above).

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The author resolved the knots labeled 1 and 2 in his 18-inch scope, knots 3–5 in his 24-inch, and the remaining three knots in Texan amateur Jimi Lowrey's 48-inch.

A side branch splits off on the northeast side and curls more tightly south, forming an inner arm halfway to the center.

A third, fainter arm emerges from the glow on the western side of the core and shoots sharply to the north. The fourth, southern arm never cleanly separates from the central glow, but a dip in brightness defines its outer edge as it sweeps west.

A number of field stars are sprinkled across the face of the galaxy, including an 11.5-magnitude luminary at the southern edge of the halo. A 20″ pair of 13th-magnitude stars lies 2.4′ southwest of the nucleus. Just 1.5′ northwest of this pair I picked up a soft, 15″ knot (labeled **2**) that could easily be missed. Studies reveal that this small-seeming patch is actually a huge stellar and gas complex spanning 2,000 light-years and containing more than a dozen tightly packed clusters, including a million-solar-mass SSC. At an age of only 10–15 million years, this supercluster is thought to be an early evolutionary stage in the formation of a classical globular star cluster.

The arms take on a clumpy texture in my 24-inch Dobsonian as more star-forming regions near the core begin to resolve. These include two slightly brighter regions (labeled **3** and **4**) along the main northern arm and an elongated enhancement at the southern tip of the inner eastern arm (**5**). Through Jimi Lowrey's 48-inch in western Texas, the view approaches the photographic appearance, with additional isolated knots in the outer halo, including **6** on the west side, **7** on the northwest end, and **8** at the north edge.

Spiral arms are challenging, low-contrast features and require the darkest possible skies, patience, and careful study. For the best view of this galaxy's overall structure, experiment with low to medium power (exit pupils from 2.5 to 4 mm). But to capture an H II region or the SSC, you'll need to increase the magnification and use the labeled image as a guide. ✦

***Steve Gottlieb** observes deep-sky objects from sites near his California home and around the world.*